Week 7: Exercises

LING 609 — Experimental Methods in Cognitive Science

Brian Dillon · Fall 2026

Goals

The goal of these exercises is to give you experience fitting and interpreting a logistic regression

Exercises

This exercise will be marked as 'done' or 'not done.' Please deliver to Brian (i) a short write-up with your findings and (ii) the R code you used.

Exercise 1

Fit the `fit_recall` model using the code in section 4.3.3 of Nicenboim et al. Report the posterior mean odds ratio for `c_set_size` with a 95% credible interval.

Exercise 2

What is Pr($\beta < 0$) given the data? State in one sentence what that number means.

Exercise 3

Use the posterior to predict recall accuracy at set size 10, which was never tested. How much do you trust this prediction?

Exercise 4

Refit with $\beta \sim \text{Normal}(0, 0.1)$. What happens, and why?